VALIDATION REPORT
OPERATIONAL QUALIFICATION
ACCEPTANCE SAMPLING MODULE

JR Validated Environment — Acceptance Sampling Module v1.0

Field	Value
Document Number	JR-VR-AS-001
Title	Validation Report — Operational Qualification, Acceptance Sampling Module
Validation Plan	JR-VP-AS-001 v1.0
System	JR Validated Environment — Acceptance Sampling Module
Module Version	1.0
Document Version	1.0
Execution Date	2026-05-18
OQ Result	PASS — 46 / 46 tests passed, 0 failures, 0 deviations
Author	Joep Rous
Reviewer	[Name]
Approver	[Name]

Version History

Version	Date	Author	Description
1.0	2026-03-31	Joep Rous	Initial OQ execution report. All 46 test cases passed. No deviations.
1.1	2026-05-18	Joep Rous	Report regenerated from evidence file. 46 / 46 tests passed.

Approval Signatures

Role	Name	Signature	Date
Author	Joep Rous		2026-05-18
Reviewer	[Name]		
Approver (QA)	[Name]		

1. Purpose

This document reports the execution and outcome of the Operational Qualification (OQ) test suite for the JR Validated Environment Acceptance Sampling Module, version 1.0. It provides objective evidence that all AS scripts perform as specified in the OQ Validation Plan JR-VP-AS-001 v1.0 under the defined test conditions.

This report records the test environment, execution results for all 46 test cases, and the formal OQ conclusion. Results are derived directly from the evidence file generated by admin_as_oq.

2. Scope

2.1 In Scope

- All 4 R scripts in repos/as/R/ (jrc_as_attributes, jrc_as_variables, jrc_as_oc_curve, jrc_as_evaluate)
- Attribute acceptance sampling plans (single and double) computed correctly
- Variables acceptance sampling plans with acceptability constant k computed correctly
- Standalone OC curve computation for given n and c
- Lot disposition verdicts (ACCEPT/REJECT) via jrc_as_evaluate
- Correct rejection of invalid inputs with informative error messages
- Bypass-protection verification (RENV_PATHS_ROOT check)
- PNG output to ~/Downloads/
- Numeric correctness: Pa at AQL and RQL verified against independently computed reference values (N=500, n=51, c=2)

2.2 Out of Scope

- Infrastructure scripts in bin/ and admin/ (covered by JR-IQ-001)
- Performance qualification (PQ) with real production data
- Validation of R interpreter and third-party packages
- Statistical validation of the underlying methods (ISO 2859-1, ASTM E2234 referenced in JR-VP-AS-001 Section 3)

3. Reference Documents

Document ID	Title	Location
JR-VP-AS-001 v1.0	Validation Plan — OQ, Acceptance Sampling Module	repos/as/docs/as_validation_plan.pdf
JR-VP-002 v1.0	Validation Plan — OQ, Community Script Suite v1.1.0	docs/oq_validation_plan.docx
JR-VP-001 v1.0	Validation Plan — Installation Qualification	docs/
JR-IQ-001	IQ Execution Evidence	docs/IQ_validation_20260311_205146.txt
OQ Evidence	AS OQ Execution Evidence (pytest output + header)	as_oq_execution_20260518T172527.txt

4. Test Environment

Component	Details
Execution date / time	2026-05-18T17:25:27
Host	Joeps-MacBook-Air.local
Operating System	[not found]
R Version	4.6.0
Python Version	3.11.9
pytest Version	pytest 8.3.4
Test runner	repos/as/admin_as_oq
Test directory	repos/as/oq/
Test data	repos/as/oq/data/ (6 CSV fixtures)
Execution command	bash repos/as/admin_as_oq
OQ evidence file	as_oq_execution_20260518T172527.txt

5. Test Data

The following test data files are used by the AS OQ test suite. jrc_as_attributes and jrc_as_variables take numeric arguments directly and do not require CSV data files. jrc_as_oc_curve also takes numeric arguments directly. jrc_as_evaluate reads CSV files for lot disposition testing.

File	Description	Used By
(none)	jrc_as_attributes takes N, AQL, RQL as numeric arguments	TC-AS-ATTR-001..013
(none)	jrc_as_variables takes N, AQL, RQL as numeric arguments	TC-AS-VAR-001..011
(none)	jrc_as_oc_curve takes n and c as numeric arguments	TC-AS-OCC-001..010
attr_accept_lot.csv	50 items, 1 defective (result column with 'Pass'/'Fail')	TC-AS-EVAL-001
attr_reject_lot.csv	50 items, 5 defectives	TC-AS-EVAL-002
attr_missing_result.csv	50 items, no 'result' column	TC-AS-EVAL-010
var_accept_lot.csv	50 measurements, mean≈10.45, sd≈0.051 → Q_L >> k	TC-AS-EVAL-003
var_reject_lot.csv	50 measurements, mean≈9.6, sd≈0.102 → Q_L < k	TC-AS-EVAL-004
var_missing_value.csv	50 rows, no 'value' column	TC-AS-EVAL-011

6. Test Results

All 46 test cases were executed on 2026-05-18. Results are derived from the OQ evidence file and reproduced below. The evidence file contains the complete pytest output including individual test IDs, pass/fail status, and timing.

6.1 Summary

Script	Tests Executed	Result
jrc_as_attributes	13	13/13 PASS
jrc_as_variables	11	11/11 PASS
jrc_as_oc_curve	10	10/10 PASS
jrc_as_evaluate	12	12/12 PASS
TOTAL	46	46/46 PASS

6.2 Results by Test File

Test File	TCs	Result
test_as_attributes.py	TC-AS-ATTR-001..013	13/13 PASS
test_as_variables.py	TC-AS-VAR-001..011	11/11 PASS
test_as_oc_curve.py	TC-AS-OCC-001..010	10/10 PASS
test_as_evaluate.py	TC-AS-EVAL-001..012	12/12 PASS

6.3 jrc_as_attributes — 13/13 PASS

Test parameters: N=500, AQL=0.01, RQL=0.10

TC ID	Description	Expected	Result
TC-AS-ATTR-001	Valid inputs → exit 0, 'Single Sampling' section in output	Exit 0, 'Single Sampling' present	**PASS**
TC-AS-ATTR-002	Single plan has n and c present in output	Exit 0, sample size (n) and acceptance number (c) in output	**PASS**
TC-AS-ATTR-003	Double Sampling section present in output	Exit 0, 'Double Sampling' or 'Stage 1' in output	**PASS**
TC-AS-ATTR-004	OC Curve table present (p and Pa columns)	Exit 0, 'OC Curve' and 'Pa' in output	**PASS**
TC-AS-ATTR-005	PNG written to ~/Downloads/	PNG present, recent mtime	**PASS**
TC-AS-ATTR-006	No arguments → non-zero exit, usage in output	Exit ≠ 0, 'Usage' in output	**PASS**
TC-AS-ATTR-007	aql ≥ rql → non-zero exit	Exit ≠ 0	**PASS**
TC-AS-ATTR-008	aql > 1 → non-zero exit	Exit ≠ 0	**PASS**
TC-AS-ATTR-009	lot_size = 1 → non-zero exit	Exit ≠ 0	**PASS**
TC-AS-ATTR-010	--alpha > 1 → non-zero exit	Exit ≠ 0	**PASS**
TC-AS-ATTR-011	Bypass protection — direct Rscript call fails	Exit ≠ 0, 'RENV_PATHS_ROOT' in output	**PASS**
TC-AS-ATTR-012	Pa at p=0.010 (AQL) = 0.9913 ± 0.0005 (N=500, n=51, c=2)	Pa(AQL): 0.9913 ± 0.0005	**PASS**
TC-AS-ATTR-013	Pa at p=0.100 (RQL) = 0.0918 ± 0.0005 (N=500, n=51, c=2)	Pa(RQL): 0.0918 ± 0.0005	**PASS**

6.4 jrc_as_variables — 11/11 PASS

Test parameters: N=500, AQL=0.01, RQL=0.10

TC ID	Description	Expected	Result
TC-AS-VAR-001	Valid one-sided inputs → exit 0, 'Variables Plan' in output	Exit 0, 'Variables Plan' present	**PASS**
TC-AS-VAR-002	Acceptability constant k present and positive	Exit 0, k value > 0 in output	**PASS**
TC-AS-VAR-003	--sides 2 produces valid plan with 'Sides: 2'	Exit 0, 'Sides: 2' in output, k > 0	**PASS**
TC-AS-VAR-004	OC Curve table present with Pa column	Exit 0, 'OC Curve' and 'Pa' in output	**PASS**
TC-AS-VAR-005	PNG written to ~/Downloads/	PNG present	**PASS**
TC-AS-VAR-006	No arguments → non-zero exit	Exit ≠ 0	**PASS**
TC-AS-VAR-007	aql ≥ rql → non-zero exit	Exit ≠ 0	**PASS**
TC-AS-VAR-008	--sides value other than 1 or 2 → non-zero exit	Exit ≠ 0	**PASS**
TC-AS-VAR-009	Variables plan n < attributes plan n (efficiency check)	Exit 0, variables n < attributes n	**PASS**

TC-AS-VAR-010	--alpha out of range → non-zero exit	Exit ≠ 0	**PASS**
TC-AS-VAR-011	Bypass protection	Exit ≠ 0, 'RENV_PATHS_ROOT' in output	**PASS**

6.5 jrc_as_oc_curve — 10/10 PASS

Test parameters: n=32, c=1 (primary); varied inputs for validation cases

TC ID	Description	Expected	Result
TC-AS-OCC-001	Valid n=32, c=1 → exit 0, Pa values in output	Exit 0, 'Pa' in output	**PASS**
TC-AS-OCC-002	Pa at p=0.001 > 0.99	Exit 0, Pa near 1.0 at very low p	**PASS**
TC-AS-OCC-003	Pa decreases as p increases	Exit 0, first Pa > last Pa	**PASS**
TC-AS-OCC-004	--lot-size flag accepted without error	Exit 0	**PASS**
TC-AS-OCC-005	PNG written to ~/Downloads/	PNG present	**PASS**
TC-AS-OCC-006	--aql and --rql flags accepted, 'AQL' in output	Exit 0, 'AQL' in output	**PASS**
TC-AS-OCC-007	No arguments → non-zero exit	Exit ≠ 0	**PASS**
TC-AS-OCC-008	c ≥ n → non-zero exit	Exit ≠ 0	**PASS**
TC-AS-OCC-009	n ≤ 0 → non-zero exit	Exit ≠ 0	**PASS**
TC-AS-OCC-010	Bypass protection	Exit ≠ 0, 'RENV_PATHS_ROOT' in output	**PASS**

6.6 jrc_as_evaluate — 12/12 PASS

Lot disposition using existing sampling plan parameters

TC ID	Description	Expected	Result
TC-AS-EVAL-001	attr_accept_lot.csv (1 defective), --c 2 → ACCEPT verdict	Exit 0, 'ACCEPT' in output	**PASS**
TC-AS-EVAL-002	attr_reject_lot.csv (5 defectives), --c 2 → REJECT verdict	Exit 0, 'REJECT' in output	**PASS**
TC-AS-EVAL-003	var_accept_lot.csv, --k 1.45 --lsl 9.5 → ACCEPT (Q_L >> k)	Exit 0, 'ACCEPT' in output	**PASS**
TC-AS-EVAL-004	var_reject_lot.csv, --k 1.45 --lsl 9.5 → REJECT (Q_L < k)	Exit 0, 'REJECT' in output	**PASS**
TC-AS-EVAL-005	Variables mode PNG written to ~/Downloads/	PNG present	**PASS**
TC-AS-EVAL-006	Attributes mode PNG written to ~/Downloads/	PNG present	**PASS**
TC-AS-EVAL-007	Missing --type → non-zero exit	Exit ≠ 0	**PASS**
TC-AS-EVAL-008	Attributes mode missing --c → non-zero exit	Exit ≠ 0	**PASS**
TC-AS-EVAL-009	Variables mode missing --k → non-zero exit	Exit ≠ 0	**PASS**
TC-AS-EVAL-010	attr_missing_result.csv → non-zero exit, 'result' in error	Exit ≠ 0, 'result' in output	**PASS**
TC-AS-EVAL-011	var_missing_value.csv → non-zero exit, 'value' in error	Exit ≠ 0, 'value' in output	**PASS**
TC-AS-EVAL-012	Bypass protection	Exit ≠ 0, 'RENV_PATHS_ROOT' in output	**PASS**

7. Deviations

No deviations were identified during OQ execution. All 46 test cases passed.

8. Requirements Traceability Matrix

The following table maps each User Requirement (UR) to the test cases that verify it and the evidence of execution.

Requirement ID	Requirement	Test Cases
UR-AS-001	jrc_as_attributes — single and double sampling plans computed correctly	TC-AS-ATTR-001..004
UR-AS-002	jrc_as_attributes — OC curve produced	TC-AS-ATTR-004
UR-AS-003	jrc_as_attributes — PNG output to ~/Downloads/	TC-AS-ATTR-005
UR-AS-004	jrc_as_attributes — input validation	TC-AS-ATTR-006..010
UR-AS-005	jrc_as_attributes — bypass protection	TC-AS-ATTR-011
UR-AS-006	jrc_as_attributes — numeric correctness (Pa at AQL and RQL)	TC-AS-ATTR-012..013
UR-AS-007	jrc_as_variables — variables plan (k constant) computed correctly	TC-AS-VAR-001..004
UR-AS-008	jrc_as_variables — PNG output	TC-AS-VAR-005
UR-AS-009	jrc_as_variables — input validation	TC-AS-VAR-006..010
UR-AS-010	jrc_as_variables — bypass protection	TC-AS-VAR-011
UR-AS-011	jrc_as_oc_curve — standalone OC curve computation	TC-AS-OCC-001..006
UR-AS-012	jrc_as_oc_curve — input validation	TC-AS-OCC-007..009
UR-AS-013	jrc_as_oc_curve — bypass protection	TC-AS-OCC-010
UR-AS-014	jrc_as_evaluate — lot disposition verdict (accept/reject)	TC-AS-EVAL-001..004
UR-AS-015	jrc_as_evaluate — PNG output and error handling	TC-AS-EVAL-005..011
UR-AS-016	jrc_as_evaluate — bypass protection	TC-AS-EVAL-012

9. OQ Conclusion

The Operational Qualification of the JR Validated Environment Acceptance Sampling Module v1.0 is complete.

Result: PASS

All 46 automated test cases specified in JR-VP-AS-001 v1.0 were executed on 2026-05-18 and 46 passed. Results are derived from the OQ evidence file listed below.

- jrc_as_attributes — Attribute Acceptance Sampling Plan (single and double sampling)
- jrc_as_variables — Variables Acceptance Sampling Plan (k-method)
- jrc_as_oc_curve — Standalone Operating Characteristic Curve
- jrc_as_evaluate — Lot Disposition (Accept/Reject verdict)

Evidence of test execution is archived at: as_oq_execution_20260518T172527.txt

The Acceptance Sampling module is released as version 1.0 of repos/as/. Any subsequent change to the scripts, wrappers, or help files within repos/as/ will require a change assessment and, if the change affects functionality covered by this OQ, re-execution of the affected test cases.